

Lab 4

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Q: - Write down the sequence for all necessary inputs for computing $4 + 5 - 7$. You will need to load number 4, 5, 7 into the Datapath, then perform the necessary calculation, and finally store the result in register number 0: -

Final Sequence of Instructions

1. $OP = 011, in = 0001, rd = 01$ (Loading $0001_2 = 4_{10}$ into the Register 1)
2. $OP = 011, in = 0101, rd = 10$ (Loading $0101_2 = 5_{10}$ into the Register 2)
3. $OP = 011, in = 0111, rd = 11$ (Loading $0111_2 = 7_{10}$ into the Register 3)
4. $OP = 110, rd = 01, rs = 10, rd = 00$ (adding reg1 with reg2 assigning the value to reg0)
5. $OP = 111, rd = 00, rs = 11, rd = 00$ (subtracting reg3 from reg0, assigning the value to reg0)